The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Germ-Anchored Exact-Shell Transport Core
Collapse to a Germ-Anchored Exact-Shell
Transport-Section Core

Abstract

The current primitive-reservoir observer-reduction frontier already sharpened the live burden inside the nucleus-generating substrate proto-germ line: the full proto-germ package was compressed to a smaller germ-anchored exact-shell transport core $\mathfrak{P}^{\mathrm{sub,proto,xtc}}$. Under the recorded routing safeguard and the benchmark one-metric upstream compression rule, however, the honest next move below that frontier is not another same-output deeper-origin relay that merely preserves that current core. The sharper admissible move is to ask whether the present transport-core presentation itself still carries benchmark-neutral redundancy.

The present note proves that it does. Relative to the already fixed current germ nucleus, the only independent proto-germ datum still carried benchmark-facingly by the current core is the canonical exact-shell transport section from the fixed germ shell into the proto shell. The exact proto shell is exactly the image of that section, the shell projection back to the fixed germ shell is its inverse, and the composite transport to the fixed seed shell factors through the same datum. Separate listing of the exact proto shell, its shell projection, and the ambient off-shell proto presentation therefore does not add independent benchmark-facing freedom at current scope.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. The positive result is therefore a genuine smaller collapse strictly inside the current transport core: the live burden now contracts to a germ-anchored exact-shell transport-section core $\mathfrak{P}^{\mathrm{sub,proto,sec}}$. What remains open is to derive or justify that sharper section core from still more primitive substrate structure, or else prove a still sharper collapse strictly inside it.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity is again constrained by a routing safeguard. Once a benchmark-facing datum has been compressed to a smaller observer-relevant core, another theorem that merely replays that same datum through a deeper relay does not count as new front-line progress unless it changes benchmark-facing status by more than depth alone. The current active-primary theorem made exactly such a compression at proto-germ scope: it removed the full nucleus-generating substrate proto-germ package as the live burden and replaced it by the smaller germ-anchored exact-shell transport core $\mathfrak{P}^{\mathrm{sub,proto,xtc}}$.

That theorem also fixes the next honest question. Inside the current transport-core presentation itself, what is genuinely independent observer-relevant data, and what is merely presentational bookkeeping once the recorded current germ nucleus is already fixed? The present note takes the

sharper internal-collapse route rather than a same-output deeper-origin relay. It shows that the exact proto shell, the restricted shell projection to the fixed germ shell, and the ambient off-shell proto presentation do not survive as independent benchmark-facing freedom once one retains the single canonical section that lifts the fixed germ shell into the proto shell.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not introduce a second operative metric, it does not enlarge the low-energy matter law, it does not change the ordinary matter route, and it does not reopen older seed, root, foundation, or earlier relay layers as if they were again the live burden. Its narrower benchmark-facing role is to isolate the only proto-germ datum still needed to present the current shell-level transport once the fixed current germ nucleus is already in hand.

2. Fixed Inputs

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}},$$

a visible reduction map

$$\mathcal{R}_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \to X_\sigma,$$

and shell-level current and equilibrium carriers

$$\mathfrak{C}_\sigma^{\mathrm{cur}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{cur}}, \qquad \mathfrak{C}_\sigma^{\mathrm{eq}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{eq}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*.*

Definition 2 (Recorded downstream seed package)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The active line already fixes the downstream seed package at current benchmark-facing scope:*

1. *a finite-dimensional seed state space* $\mathcal{S}_\sigma^{\mathrm{seed}}$*, a compact convex seed body* $\mathcal{B}_\sigma^{\mathrm{seed}} \subset \mathcal{S}_\sigma^{\mathrm{seed}}$*, the exact seed shell* $\mathcal{Z}_\sigma^{\mathrm{seed}} \subset \mathcal{B}_\sigma^{\mathrm{seed}}$*, and the exact seed support shell* $\mathcal{U}_\sigma^{\mathrm{seed,sup}} \subset \mathcal{Z}_\sigma^{\mathrm{seed}}$*;*

2. *a seed zero core* $\mathcal{Z}_\sigma^{0,\mathrm{seed}} \subset \mathcal{U}_\sigma^{\mathrm{seed,sup}}$*, the same hidden fiber space* $\mathcal{H}_\sigma^{\mathrm{hid}}$ *with compact hidden body* $D_\sigma^{\mathrm{hid}} \subset \mathcal{H}_\sigma^{\mathrm{hid}}$*, and the fixed seed split*
$$\Phi_\sigma^{\mathrm{seed}} : \mathcal{Z}_\sigma^{0,\mathrm{seed}} \times D_\sigma^{\mathrm{hid}} \xrightarrow{\cong} \mathcal{U}_\sigma^{\mathrm{seed,sup}};$$

3. *the seed hidden charge*
$$\mathcal{Q}_\sigma^{\mathrm{seed}} : \mathcal{H}_\sigma^{\mathrm{hid}} \to Q_\sigma^{\mathrm{eqn}},$$
a smooth observer-compatible exact seed zero-response realization
$$\zeta_\sigma^{\mathrm{seed}} : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{Z}_\sigma^{0,\mathrm{seed}},$$
and the fixed-data solvability / coercive package recorded through $\widetilde{K}_\sigma^{0,\mathrm{seed}}$ *and* $P_\sigma^{\mathrm{seed,co}}$*;*

4. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, and no premature promotion from adoption to completed derivation.*

Definition 3 (Recorded current minimal germ zero-shell transport nucleus)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The recorded current minimal germ zero-shell transport nucleus* $\mathfrak{G}^{\mathrm{sub,germ,zstn}}$ *for sector* σ *consists of:*

1. *the germ state space* $\mathcal{G}_\sigma^{\mathrm{germ}}$*, compact convex germ body* $\mathcal{B}_\sigma^{\mathrm{germ}} \subset \mathcal{G}_\sigma^{\mathrm{germ}}$*, projection*
$$\Pi_\sigma^{\mathrm{germ}} : \mathcal{G}_\sigma^{\mathrm{germ}} \to \mathcal{S}_\sigma^{\mathrm{seed}},$$
and seed section
$$\sigma_\sigma^{\mathrm{germ}\leftarrow\mathrm{seed}} : \mathcal{B}_\sigma^{\mathrm{seed}} \to \mathcal{B}_\sigma^{\mathrm{germ}}$$
satisfying
$$\Pi_\sigma^{\mathrm{germ}}(\mathcal{B}_\sigma^{\mathrm{germ}}) = \mathcal{B}_\sigma^{\mathrm{seed}}, \qquad \Pi_\sigma^{\mathrm{germ}} \circ \sigma_\sigma^{\mathrm{germ}\leftarrow\mathrm{seed}} = \mathrm{id}_{\mathcal{B}_\sigma^{\mathrm{seed}}};$$

2. *the restriction section*
$$\Gamma_\sigma^{\mathrm{germ}} : \mathcal{G}_\sigma^{\mathrm{germ}} \to E_\sigma^{\mathrm{sub}},$$
and the exact germ shell
$$\mathcal{Z}_\sigma^{\mathrm{germ}} \equiv \left(\Gamma_\sigma^{\mathrm{germ}}\right)^{-1}(0) \cap \mathcal{B}_\sigma^{\mathrm{germ}},$$
with
$$\Pi_\sigma^{\mathrm{germ}}\big|_{\mathcal{Z}_\sigma^{\mathrm{germ}}} : \mathcal{Z}_\sigma^{\mathrm{germ}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{seed}};$$

3. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, and no premature promotion from adoption to completed derivation.*

By the already-recorded current frontier theorem, the support shell, zero core, hidden split, hidden charge, exact zero-response realization, and fixed-data solvability / coercive package are then canonically induced downstream from this nucleus together with the fixed seed package of Definition 2, rather than taken as additional independent benchmark-facing data.

Remark 1. *Definitions 1, 2, and 3 are fixed inputs. The primitive continuation data, the one operative metric, the ordinary matter route, the recorded downstream seed package, and the recorded current germ nucleus are not reopened here. The present note asks only which part of the current germ-anchored proto transport presentation still carries independent benchmark-facing burden once that current germ nucleus is already fixed.*

Remark 2 (Downstream fixed fact)**.** *The already-recorded current frontier theorem proves that once a minimal germ zero-shell transport nucleus is fixed, the larger observer-relevant germ seed-transport core is canonically recovered and carries no extra benchmark-relevant freedom. The present note uses that downstream fact only as a routing consequence: after the current germ nucleus is held fixed, no larger current-core package needs to be postulated anew.*

3. Current Germ-Anchored Exact-Shell Transport Core

Definition 4 (Germ-anchored exact-shell transport core)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. The* germ-anchored exact-shell transport core $\mathfrak{P}^{\mathrm{sub,proto,xtc}}$ *for sector* σ *consists of:*

1. *the proto-germ state space* $\mathcal{P}_\sigma^{\mathrm{proto}}$*, compact convex proto-germ body* $\mathcal{B}_\sigma^{\mathrm{proto}} \subset \mathcal{P}_\sigma^{\mathrm{proto}}$*, and affine surjection*
$$\Pi_\sigma^{\mathrm{proto}} : \mathcal{P}_\sigma^{\mathrm{proto}} \to \mathcal{G}_\sigma^{\mathrm{germ}}$$
satisfying
$$\Pi_\sigma^{\mathrm{proto}}(\mathcal{B}_\sigma^{\mathrm{proto}}) = \mathcal{B}_\sigma^{\mathrm{germ}};$$
2. *the restriction section*
$$\Gamma_\sigma^{\mathrm{proto}} : \mathcal{P}_\sigma^{\mathrm{proto}} \to E_\sigma^{\mathrm{sub}},$$
the exact proto-germ shell
$$\mathcal{Z}_\sigma^{\mathrm{proto}} \equiv (\Gamma_\sigma^{\mathrm{proto}})^{-1}(0) \cap \mathcal{B}_\sigma^{\mathrm{proto}},$$
and the exact-shell transport diffeomorphism
$$\Pi_\sigma^{\mathrm{proto}}|_{\mathcal{Z}_\sigma^{\mathrm{proto}}} : \mathcal{Z}_\sigma^{\mathrm{proto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{germ}};$$
3. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

Remark 3. *Definition 4 is the current benchmark-facing frontier before the present theorem is applied. The previous active-primary theorem already removed the full nucleus-generating substrate proto-germ package as the live burden and showed that the omitted full-body proto lift is benchmark neutral. The present note asks whether the current transport-core presentation is itself still larger than necessary once the recorded current germ nucleus is already fixed downstream.*

4. Germ-Anchored Exact-Shell Transport-Section Core

Definition 5 (Germ-anchored exact-shell transport-section core)**.** *Fix $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ and a benchmark-faithful germ-anchored exact-shell transport core in the sense of Definition 4. The* germ-anchored exact-shell transport-section core $\mathfrak{P}^{\mathrm{sub,proto,sec}}$ *for sector σ consists of:*

1. *the canonical exact-shell transport section*

$$\mathfrak{s}_\sigma^{\mathrm{proto,sec}} \equiv \left(\Pi_\sigma^{\mathrm{proto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{proto}}}\right)^{-1} : \mathcal{Z}_\sigma^{\mathrm{germ}} \to \mathcal{Z}_\sigma^{\mathrm{proto}} \subset \mathcal{B}_\sigma^{\mathrm{proto}} \subset \mathcal{P}_\sigma^{\mathrm{proto}};$$

2. *benchmark faithfulness in the sense of Definition 4.*

The exact proto shell and the restricted shell projection back to $\mathcal{Z}_\sigma^{\mathrm{germ}}$ are understood as derived data rather than separately listed core data.

Remark 4. *Definition 5 is strictly smaller than Definition 4 at current scope. Once the current germ shell $\mathcal{Z}_\sigma^{\mathrm{germ}}$ is already fixed upstream, the exact proto shell and its shell projection are recovered from the single canonical section $\mathfrak{s}_\sigma^{\mathrm{proto,sec}}$; they need not be listed independently as live benchmark-facing data.*

5. Collapse of the Current Transport Core

Proposition 1 (The canonical transport section reconstructs the exact proto shell)**.** *For every benchmark-faithful germ-anchored exact-shell transport core,*

$$\mathcal{Z}_\sigma^{\mathrm{proto}} = \mathfrak{s}_\sigma^{\mathrm{proto,sec}}\big(\mathcal{Z}_\sigma^{\mathrm{germ}}\big),$$

and the section

$$\mathfrak{s}_\sigma^{\mathrm{proto,sec}} : \mathcal{Z}_\sigma^{\mathrm{germ}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{proto}}$$

is a diffeomorphism with inverse

$$\Pi_\sigma^{\mathrm{proto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{proto}}} : \mathcal{Z}_\sigma^{\mathrm{proto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{germ}}.$$

Proof. By Definition 5(1),

$$\mathfrak{s}_\sigma^{\mathrm{proto,sec}} = \left(\Pi_\sigma^{\mathrm{proto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{proto}}}\right)^{-1}.$$

Therefore

$$\mathfrak{s}_\sigma^{\mathrm{proto,sec}}\big(\mathcal{Z}_\sigma^{\mathrm{germ}}\big) = \mathcal{Z}_\sigma^{\mathrm{proto}}.$$

Because Definition 4(2) already states that

$$\Pi_\sigma^{\mathrm{proto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{proto}}} : \mathcal{Z}_\sigma^{\mathrm{proto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{germ}}$$

is a diffeomorphism, its inverse $\mathfrak{s}_\sigma^{\mathrm{proto,sec}}$ is also a diffeomorphism, and the displayed inverse relation follows immediately. □

Proposition 2 (The composite transport to the fixed seed shell factors through the section core)**.** *For every benchmark-faithful germ-anchored exact-shell transport core, the composite shell transport*

to the fixed seed shell is reconstructed from the section core by

$$\Pi_\sigma^{\mathrm{germ}}\big|_{\mathcal{Z}_\sigma^{\mathrm{germ}}} \circ \left(\mathfrak{s}_\sigma^{\mathrm{proto,sec}}\right)^{-1} : \mathcal{Z}_\sigma^{\mathrm{proto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{seed}}.$$

Proof. Proposition 1 gives

$$\left(\mathfrak{s}_\sigma^{\mathrm{proto,sec}}\right)^{-1} = \Pi_\sigma^{\mathrm{proto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{proto}}} : \mathcal{Z}_\sigma^{\mathrm{proto}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{germ}}.$$

Definition 3(2) gives the fixed shell diffeomorphism

$$\Pi_\sigma^{\mathrm{germ}}\big|_{\mathcal{Z}_\sigma^{\mathrm{germ}}} : \mathcal{Z}_\sigma^{\mathrm{germ}} \xrightarrow{\cong} \mathcal{Z}_\sigma^{\mathrm{seed}}.$$

Composing these two diffeomorphisms yields the displayed transport from $\mathcal{Z}_\sigma^{\mathrm{proto}}$ to $\mathcal{Z}_\sigma^{\mathrm{seed}}$. □

Proposition 3 (All current benchmark-facing uses of the transport core factor through the section core)**.** *For every benchmark-faithful germ-anchored exact-shell transport core, the benchmark-facing role of the current transport core on the active line is already carried by the section core* $\mathfrak{P}^{\mathrm{sub,proto,sec}}$ *together with the fixed current germ nucleus* $\mathfrak{G}^{\mathrm{sub,germ,zstn}}$*. In particular, no additional current benchmark-facing consequence requires more than the canonical section* $\mathfrak{s}_\sigma^{\mathrm{proto,sec}}$*, the fixed current germ nucleus, and benchmark faithfulness.*

Proof. Proposition 1 reconstructs the exact proto shell and its transport back to the fixed germ shell from the canonical section $\mathfrak{s}_\sigma^{\mathrm{proto,sec}}$. Proposition 2 then reconstructs the composite shell transport to the fixed seed shell. The recorded current germ nucleus of Definition 3 already fixes the germ body, the seed anchor, the germ restriction section, and the exact germ shell at benchmark-facing scope; those data are not altered by changing the presentation of the proto package outside the lifted shell. By Remark 2, once that current germ nucleus is fixed, the downstream observer-relevant germ seed-transport core is already canonically recovered and carries no additional independent benchmark-facing freedom. Therefore every current benchmark-facing use of the transport core factors through the smaller section core together with the already fixed current germ nucleus. □

Proposition 4 (Separate shell presentation and ambient proto presentation are benchmark neutral at current scope)**.** *For every benchmark-faithful germ-anchored exact-shell transport core, the separate presentation of*

$$\mathcal{Z}_\sigma^{\mathrm{proto}}, \qquad \Pi_\sigma^{\mathrm{proto}}\big|_{\mathcal{Z}_\sigma^{\mathrm{proto}}}, \qquad \mathcal{B}_\sigma^{\mathrm{proto}},$$

and the off-shell restriction-section presentation of $\Gamma_\sigma^{\mathrm{proto}}$ *away from*

$$\mathfrak{s}_\sigma^{\mathrm{proto,sec}}(\mathcal{Z}_\sigma^{\mathrm{germ}})$$

does not add independent benchmark-facing freedom beyond the section core $\mathfrak{P}^{\mathrm{sub,proto,sec}}$*.*

Proof. By Proposition 1, the exact proto shell and the restricted shell projection are reconstructed from the canonical section $\mathfrak{s}_\sigma^{\mathrm{proto,sec}}$. By Proposition 3, all current benchmark-facing uses of the transport core on the active line already factor through that section together with the fixed current germ nucleus. Therefore the separate listing of the exact proto shell and its shell projection does not add independent benchmark-facing freedom.

The remaining data in Definition 4 present an ambient proto-germ extension of that lifted shell inside $\mathcal{P}_\sigma^{\mathrm{proto}}$. At current benchmark-facing scope, however, the active line uses the proto package only to transport the fixed germ shell faithfully while preserving the already fixed one-metric observer outputs. Once the canonical section has fixed that lifted shell transport and benchmark faithfulness is retained, changing the ambient off-shell proto presentation does not alter the fixed germ nucleus, the downstream observer-relevant germ seed-transport core, the operative metric $g^\sharp$, or the ordinary matter route. Hence those ambient presentation choices are benchmark neutral at current scope. □

Proposition 5 (The benchmark package remains fixed)**.** *Every germ-anchored exact-shell transport-section core preserves the same benchmark one-metric package* (1)*. In particular, the operative metric remains exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

Proof. This is part of benchmark faithfulness in Definition 5(2). The present note only removes presentational redundancy inside the current transport core. It does not alter the benchmark exact-closure package. □

6. Main Theorem

Theorem 1 (Germ-anchored exact-shell transport core collapse to a germ-anchored exact-shell transport-section core)**.** *Fix the primitive continuation data of Definition 1, the recorded downstream seed package of Definition 2, the recorded current germ nucleus of Definition 3, and a benchmark-faithful germ-anchored exact-shell transport core in the sense of Definition 4. Then, for each sector $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$, the live benchmark-facing burden inside the current transport-core frontier collapses from the germ-anchored exact-shell transport core $\mathfrak{P}^{\mathrm{sub,proto,xtc}}$ to the smaller germ-anchored exact-shell transport-section core $\mathfrak{P}^{\mathrm{sub,proto,sec}}$. More precisely:*

1. *the exact proto shell is exactly the image of the canonical transport section and the restricted shell projection is its inverse by Proposition 1;*
2. *the composite shell transport to the fixed seed shell factors through that same section by Proposition 2;*
3. *all current benchmark-facing uses of the transport core on the active line already factor through the section core together with the fixed current germ nucleus by Proposition 3;*
4. *the separate shell presentation and the ambient off-shell proto presentation add no independent benchmark-facing freedom beyond the section core by Proposition 4;*
5. *the benchmark boundary remains fixed by Proposition 5;*
6. *consequently the remaining honest burden below the previous transport-core frontier is no longer the full germ-anchored exact-shell transport core $\mathfrak{P}^{\mathrm{sub,proto,xtc}}$ as such. What remains is to derive or justify the sharper section core $\mathfrak{P}^{\mathrm{sub,proto,sec}}$ from still more primitive substrate structure, or else prove a still smaller theorem-level collapse, sharper necessity result, or sharper uniqueness result strictly inside that section core.*

Proof. Item (1) is Proposition 1. Item (2) is Proposition 2. Item (3) is Proposition 3. Item (4) is Proposition 4. Item (5) is Proposition 5. Item (6) is the routing consequence of the previous items

together with the active safeguard against counting another same-output deeper-origin relay as new front-line progress when the live benchmark-relevant datum is unchanged. □

Corollary 1 (What must be done next)**.** *After Theorem 1, the next honest benchmark-facing manuscript must either:*

1. *derive or justify the germ-anchored exact-shell transport-section core* $\mathfrak{P}^{\mathrm{sub,proto,sec}}$ *from still more primitive substrate structure in a way that changes benchmark-facing status;*
2. *identify a genuinely smaller theorem-level observer-relevant datum strictly inside* $\mathfrak{P}^{\mathrm{sub,proto,sec}}$ *and prove a sharper collapse to it;*
3. *do either of the previous two while preserving the same benchmark one-metric package, the same ordinary-matter route, and the same claim boundary.*

Another same-output deeper-origin relay that preserves this section core, another re-expansion back to the larger transport-core presentation as if its recovered shell data or ambient off-shell proto presentation were still independently live, another replay of the current minimal germ zero-shell transport nucleus as if it were again the unexplained stopping point, another reopening of older seed, root, foundation, ground, origin, precursor, lift, shell-restriction, selector-law, combined-response, ambient-package, or exact-shell-affine-nucleus layers, or any move that introduces a second operative metric, enlarges the ordinary matter law, changes the ordinary matter route, or uses stronger promotion language does not satisfy the present burden.

Proof. Theorem 1 shows that the transport-core presentation is no longer the minimal benchmark-facing datum at this stage. Therefore a subsequent manuscript must improve on the smaller section core rather than merely restating the larger transport-core presentation or replaying the already-fixed current germ nucleus. □

7. Conclusion

The positive result of this note is a disciplined compression step inside the current proto-germ frontier. The previous active-primary theorem already showed that the full nucleus-generating substrate proto-germ package is not the minimal benchmark-facing datum, because its live burden collapses to a smaller germ-anchored exact-shell transport core. The present theorem then takes the honest next internal route available there: it shows that even that current transport-core presentation is still larger than necessary.

What survives as the live burden is the germ-anchored exact-shell transport-section core. Relative to the already fixed current germ shell, that sharper datum is just the canonical exact-shell transport section lifting $\mathcal{Z}_\sigma^{\mathrm{germ}}$ into the proto shell. The exact proto shell is its image, the shell projection back to the fixed germ shell is its inverse, and all current benchmark-facing uses of the transport core factor through that same section together with the already fixed current germ nucleus.

This is the honest next burden after the present theorem. A new primary manuscript must derive or justify that sharper section core from still more primitive substrate structure, or else collapse it further, while preserving the same benchmark one-metric package, the same ordinary-matter route, and the same claim boundary. The current result does not yet complete a first-principles derivation of GR from substrate microphysics, but it does narrow the live derivational burden one level further on the active line.